

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Differential Equation & Linear Algebra

Time: 1.5 Hours

Semester / Programme: First Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) Find the integrating factor of the first-order linear differential equation $x \frac{dy}{dx} + yx = 1$. [1]

$$I.F. = e^{\int P(x)dx}$$

1(b) Determine the Wronskian $W(y_1, y_2)$ of the two solutions $y_1 = e^{2x}$ and $y_2 = e^{-3x}$ of a second-order linear differential equation. [1]

$$\text{Formula: } W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

1(c) State the condition for exactness of a first-order ordinary differential equation $M(x,y)dx + N(x,y)dy = 0$. [1]

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

1(d) Find the recurrence relation for the Gamma function $\Gamma(x)$ for $x > 0$. [1]

$$\Gamma(x+1) = x\Gamma(x)$$

1(e) Write the auxiliary (characteristic) equation for the second-order differential equation $y'' - 5y' + 6y = 0$. [1]

$$ar^2 + br + c = 0$$

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Solve the Bernoulli differential equation: $(\frac{dy}{dx} + (1/x)y = xy^2)$. [2.5]

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

(b) A thermometer reading $70^\circ F$ is placed in an oven preheated to a constant temperature. Through Newton's Law of Cooling/Heating, if the thermometer reads $110^\circ F$ after 1 minute and $140^\circ F$ after 2 minutes, what is the oven temperature? [2.5]

$$\frac{dT}{dt} = -k(T - T_{oven})$$

Q.3. (a) Solve the exact differential equation: $(3x^{2y} + y^3)dx + (x^3 + 3xy^2)dy = 0$. [2.5]

$$\int_{y=\text{const}} Mdx + \int (\text{terms of } N \text{ not containing } x)dy = C$$

(b) Find the orthogonal trajectories of the family of curves $y = Cx^2$, where C is an arbitrary parameter. [2.5]

$$\text{Formula: } \left| \frac{dy}{dx} \right|_{\text{orthogonal}} = - \frac{1}{\left| \frac{dy}{dx} \right|}$$

Q.4. (a) Solve the second-order homogeneous differential equation with constant coefficients: $y'' - 4y' + 4y = 0$. [2.5]

$$y(x) = (C_1 + C_2 x)e^{rx}$$

(b) Solve the Euler-Cauchy equation: $x^2 y'' - 3x y' + 4y = 0$. [2.5]

$$m(m-1) + \alpha m + \beta = 0$$

Q.5. (a) Evaluate the Beta integral representation relation and prove that $B(m, n) = ((m)(n) / (m+n))$ for $m, n > 0$. [2.5]

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

(b) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} (3^n / n!) x^n$. [2.5]

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

--- END OF PAPER ---